

Patrick Iglesias-Zemmour

Former Researcher, CNRS (France) — 40-year career

Visiting Professor, The Hebrew University of Jerusalem, Israel

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Born 27 December 1953, French & Israeli citizenship

Research Profile. Founder of modern diffeology — a geometric framework extending differential geometry to singular and infinite-dimensional spaces.

Building on the axiomatic foundations laid by J.-M. Souriau (1980) and the category of *espaces différentiels* introduced with P. Donato (1983), I have since developed diffeology into a mature field: fiber bundles, homotopy theory, the chain homotopy operator, the moment map (AMS Memoir 2010), orbifolds, quasifolds, the bridge to noncommutative geometry (JNCG 2018, 2021), geometric quantization by paths (AMS Memoir under review), and the definitive texts *Diffeology* (AMS 2013, Beijing 2022), *Lectures on Diffeology* (2025), and the forthcoming three-volume *The Geometry of Motion*.

My work establishes systematic bridges between diffeology and:

- **Noncommutative Geometry** — Two papers in the *Journal of Noncommutative Geometry* (2018, 2021) show that C^* -algebras of singular spaces arise naturally from diffeological groupoids.
- **Geometric Quantization** — Forthcoming AMS memoir *Geometric Quantization by Paths* constructs the *Prequantum Groupoid*, unifying Feynman's path integral with the Dirac-Kostant-Souriau program.
- **Homotopy Theory** — Sustained collaboration with the Japanese school (Iwase, Kuribayashi, Shimakawa) through five editions of the *Building Differential Homotopy Theory* conference series.

Unifying theme: Universal models for singular symplectic spaces—from infinite quasi-projective spaces (Proc. AMS, 2016) to toric quasifolds (with E. Prato, JNCG, 2021) to geometric quantization by paths: the prequantum groupoid, and noncommutative geometry as the "Threshold of Analysis" (AMS memoir, under review).

Professional Positions.

Visiting Professor at Università degli Studi di Firenze Sept. 2026.

Visiting Professor, The Hebrew University of Jerusalem, Israel	2021–present
Researcher, Institut de Mathématiques de Marseille, CNRS, France	2011–2020
Visiting Professor, The Hebrew University of Jerusalem, Israel	2004–2011
Researcher, LATP, CNRS Marseille, France	1998–2004
Researcher, UMPA, École Normale Supérieure de Lyon, France	1992–1998
Researcher, Centre de Physique Théorique, CNRS Luminy, Marseille, France	1980–1992

Education.

Doctor of Science (Doctorat d'État), 1985 1985
 Université de Provence. Dissertation: *Fibrations difféologiques et homotopie*. Advisor: J.-M. Souriau.
 Board: J. Dixmier (President), D. Bennequin, J. Breuneval, P. Donato, J. Pradines, J.-M. Souriau, R. Stora.
With distinction.

PhD (Doctorat de 3ème cycle), 1979 1979
 Université de Provence. Dissertation: *Thermodynamique géométrique appliquée aux configurations tournantes en astrophysique*. Advisor: J.-M. Souriau. *With distinction.*

DEA Mathématiques Pures (Master), 1978 1978
 Université de Provence. *With merit.*

DEA Physique Théorique (Master), 1977 1977
 Université de Provence. *Valedictorian, with distinction.*

Recent Teaching Experience.

Invited Lecture Series Oct 2020 – Feb 2021
Diffeology: From Basics to Recent Developments (17 weeks, 34 hours), Shantou University, China (via Zoom). Program: "Famous Teachers Overseas" (海外名师). Materials formed the basis for the first chapters of *Lectures on Diffeology* (Beijing WPC, 2025)

Post-Graduate Course on Diffeology Apr 13 – May 3, 2026
 Tel-Aviv University, Israel.

Summer School, Turkey 2018
Two-week lecture series on diffeology, Nesin Mathematics Village.

Books and Monographs.

- 1 (2026). *The Geometry of Motion (3 volumes)*. Beijing World Publishing Corp., under contract. Preprint: <http://math.huji.ac.il/~piz/documents/TGOM.pdf>
- 2 (2025). *Lectures on Diffeology*. Beijing World Publishing Corp., 378 pages. ISBN 978-7-5232-1841-9
- 3 (2022). *Diffeology (Revised Second Edition)*. Beijing World Publishing Corp., 529 pages. ISBN 978-7-5192-9608-7
- 4 (2018). *La Géométrie des Mouvements*. Amazon Desktop Kindle Publishing, 71 pages. ISBN 978-1-9810-3206-8
- 5 (2013). *Diffeology*. American Mathematical Society, Mathematical Surveys and Monographs, vol. 185, 450 pages. ISBN 978-0-8218-9131-5
- 6 (2010). *The Moment Maps in Diffeology*. American Mathematical Society, Memoirs, no. 972, vol. 207, 80 pages. ISBN 978-0-8218-4709-1
- 7 (2000). *Symétries et Moment*. Hermann, Paris, 200 pages. ISBN 978-2-7056-6419-0

Works in Progress.

2026, *Geometric Quantization by Paths*, AMS Memoir, submitted. Based on three preprints:

<http://math.huji.ac.il/~piz/documents/GQBP-I.pdf> (Part I: Simply Connected, 2025)

<http://math.huji.ac.il/~piz/documents/GQBP-II.pdf> (Part II: General Case, 2025)

<http://math.huji.ac.il/~piz/documents/GQBP-III.pdf> (Part III: Metaplectic Anomaly, 2026)

arXiv:2508.11337, 2512.24627, 2601.23259

Selected Journal Publications.

2026, *The Boman Paradox: When Topology Plus Algebra Do Not Define Geometry*, Math. Intelligencer (2026). DOI: 10.1007/s00283-025-10500-3

2025, *Why Diffeology*, Arnold Mathematical Journal, Vol. 12, Issue 2 (DOI not yet active)

2025, *Groupoids in Diffeology*, Electronic Research Archive, Vol. 33, No. 12, pp. 7957–7973. DOI: 10.3934/era.2025350

2025, *On the Diffeology of Orbit Spaces (with S. Gürer)*, Differential Geometry and its Applications Volume 103, June 2026. DOI: 10.1016/j.difgeo.2026.102391

2024, *Editorial note to: On the motion of spinning particles in general relativity by Jean-Marie Souriau (with T. Damour)*, General Relativity and Gravitation, 56(10):127

2023, *Orbifolds as Stratified Diffeologies (with S. Gürer)*, Differential Geometry and its Applications, vol. 86

2023, *Čech–De Rham Bicomplex in Diffeology*, Israel Journal of Mathematics

2021, *Quasifolds, Diffeology and Noncommutative Geometry (with E. Prato)*, Journal of Noncommutative Geometry, vol. 15, no. 2, pp. 735–759

2021, *Every Symplectic Manifold is a (Linear) Coadjoint Orbit (with P. Donato)*, Bulletin of the Canadian Mathematical Society, pp. 1–16

2019, *Refraction and Reflexion According to the Principle of General Covariance*, Journal of Geometry and Physics, vol. 142

2018, *Noncommutative Geometry and Diffeology: The Case of Orbifolds (with J.-P. Laffineur)*, Journal of Noncommutative Geometry, vol. 12, no. 4, pp. 1551–1572

2016, *Example of Singular Reduction in Symplectic Diffeology*, Proceedings of the American Mathematical Society, vol. 144, pp. 1309–1324

2015, *Primary Spaces, Mackey's Obstruction, and the Generalized Barycentric Decomposition (with F. Ziegler)*, Journal of Symplectic Geometry, vol. 13, no. 1, pp. 55–80

2013, *Variation of Integrals in Diffeology*, Canadian Journal of Mathematics, vol. 65, no. 6, pp. 1255–1286

2011, *Smooth Lie Group Actions are Diffeological Subgroups (with Y. Karshon)*, Proceedings of the American Mathematical Society, vol. 140, pp. 731–739

2010, *Orbifolds as Diffeologies (with Y. Karshon and M. Zadka)*, Transactions of the American Mathematical Society, vol. 362, no. 6, pp. 2811–2831

2007, *Dimension in Diffeology*, Indagationes Mathematicae, vol. 18, no. 4

- 1998, *Les origines du calcul symplectique chez Lagrange*, L'Enseignement Mathématique, vol. 44, pp. 257–277
- 1995, *La trilogie du moment*, Annales de l'Institut Fourier, vol. 45, fasc. 3
- 1991, *Les $SO(3)$ -variétés symplectiques et leur classification en dimension 4*, Bulletin de la Société Mathématique de France, vol. 119, fasc. 4
- 1985, *Exemple de groupes différentiels: flots irrationnels sur le tore (with P. Donato)*, Comptes Rendus de l'Académie des Sciences, t. 301, série I, n° 4

Recent Preprints (Under Review).

- 2026, *A Conformal Co-Symplectic Structure on the Space of Pseudo-Riemannian Geodesics*, Submitted to the Journal of Symplectic Geometry. <http://math.huji.ac.il/~piz/documents/ACSSOSOPRG.pdf>
- 2025, *Does the motor cortex draw on a wire plane?*, Submitted to Biological Cybernetics. <http://math.huji.ac.il/~piz/documents/DTMCDOTWP.pdf>
- 2025, *Bosonic and Fermionic Singularities in Diffeology*, <http://math.huji.ac.il/~piz/documents/BAFSID.pdf>
- 2025, *Diffeology and Arithmetics of Irrational Tori*, <http://math.huji.ac.il/~piz/documents/DAAQIT.pdf>

Major International Collaborations.

- **Japan:** Norio Iwase (Kyushu), Katsuhiko Kuribayashi (Shinshu), Kazuhisa Shimakawa (Okayama), Hiroshi Kihara (Aizu). Invited participant in all five *Building Differential Homotopy Theory* conferences (2019–2026). Iwase's JSPS grant explicitly identifies me as the “central figure” in diffeology research.
- **Italy:** Elisa Prato (Firenze). Collaboration on quasifolds and noncommutative geometry since 2017; joint JNCG paper (2021).
- **Turkey:** Serap Gürer (Galatasaray). Collaboration since 2015; multiple joint papers on diffeology, orbifolds, and differential forms.
- **Israel/Canada:** Yael Karshon (Toronto). Collaboration since 2002; joint work on orbifolds and Lie group actions.
- **China:** Enxin Wu (Shantou). Invited lecture series (2020–21) and ongoing collaboration.

Invited Lectures (Last 5 Years).

Building Differential Homotopy Theory, Matsumoto, Japan	Mar 2026
“Geometric Quantization by Paths” and “The Boman Paradox”	
Shantou University, China	Mar 2025
“When Diffeology Meets Noncommutative Geometry”	
Duality Seminar, University of Warsaw	Mar 2025
“Why Diffeology”	
Fields Institute, Toronto, Canada	Jul 2024

“Symplectic Diffeology” (Workshop on Hamiltonian Geometry and Quantization) Building Differential Homotopy Theory, Osaka, Japan	<i>Mar 2024</i>
“Symplectic Reduction in Diffeology with Singularities” Building Differential Homotopy Theory, Aizu, Japan	<i>Mar 2023</i>
“Symplectic Diffeology” Séminaire Interactions Fondamentales, CPT Luminy	<i>Dec 2022</i>
“Développements récents en diffeologie” Università degli Studi di Firenze, Italy	<i>Oct 2022</i>
“Every Symplectic Manifold is a Linear Coadjoint Orbit”	

Invited Lectures (Earlier).

Building Differential Homotopy Theory, Matsumoto, Japan	<i>Mar 2020</i>
Series of lectures: “Diffeology” (invited course)	

Online Presence and Community Building.

- **Website:** Founder, [Diffeology.net](https://diffeology.net) – a central hub for the diffeology community.
- **Online Seminar:** Co-organizer, Monthly Global Seminar on Diffeology (2021–present).
- **Repository:** Creator of the *Diffeology Archives* on GitHub – a definitive, version-controlled, machine-readable repository of foundational works in diffeology, designed for researchers and AI training.

Conference and Workshop Organization.

- International Conference Souriau 2019, Paris Diderot University (co-organizer)
- Workshop on Diffeology, Aix-en-Provence/Istanbul (2013–2016, series organizer)
- “Nouvelles géométries pour la physique”, FRUMAM, Aix-en-Provence (2013)
- Souriau’90, International Conference in honor of J.-M. Souriau, Aix-en-Provence (2012)
- Summer School “Symétries et application moment”, CIRM Marseille (1999)
- Le Séminaire des élèves, École Normale Supérieure de Lyon (1992–1998)
- Founder and Editor-in-Chief, *Journal de Maths des Élèves de l’ENS-Lyon* (1994–1998)

Languages. French (native), English (fluent), Hebrew (working knowledge)